

NAME

setxattr, lsetxattr, fsetxattr – set an extended attribute value

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/xattr.h>

int setxattr(const char *path, const char *name,
             const void value[.size], size_t size, int flags);
int lsetxattr(const char *path, const char *name,
             const void value[.size], size_t size, int flags);
int fsetxattr(int fd, const char *name,
             const void value[.size], size_t size, int flags);
```

DESCRIPTION

Extended attributes are *name:value* pairs associated with inodes (files, directories, symbolic links, etc.). They are extensions to the normal attributes which are associated with all inodes in the system (i.e., the **stat(2)** data). A complete overview of extended attributes concepts can be found in **xattr(7)**.

setxattr() sets the *value* of the extended attribute identified by *name* and associated with the given *path* in the filesystem. The *size* argument specifies the size (in bytes) of *value*; a zero-length value is permitted.

lsetxattr() is identical to **setxattr()**, except in the case of a symbolic link, where the extended attribute is set on the link itself, not the file that it refers to.

fsetxattr() is identical to **setxattr()**, only the extended attribute is set on the open file referred to by *fd* (as returned by **open(2)**) in place of *path*.

An extended attribute name is a null-terminated string. The *name* includes a namespace prefix; there may be several, disjoint namespaces associated with an individual inode. The *value* of an extended attribute is a chunk of arbitrary textual or binary data of specified length.

By default (i.e., *flags* is zero), the extended attribute will be created if it does not exist, or the value will be replaced if the attribute already exists. To modify these semantics, one of the following values can be specified in *flags*:

XATTR_CREATE

Perform a pure create, which fails if the named attribute exists already.

XATTR_REPLACE

Perform a pure replace operation, which fails if the named attribute does not already exist.

RETURN VALUE

On success, zero is returned. On failure, *-1* is returned and *errno* is set to indicate the error.

ERRORS**EDQUOT**

Disk quota limits meant that there is insufficient space remaining to store the extended attribute.

EEXIST

XATTR_CREATE was specified, and the attribute exists already.

ENODATA

XATTR_REPLACE was specified, and the attribute does not exist.

ENOSPC

There is insufficient space remaining to store the extended attribute.

ENOTSUP

The namespace prefix of *name* is not valid.

ENOTSUP

Extended attributes are not supported by the filesystem, or are disabled,

EPERM

The file is marked immutable or append-only. (See **ioctl_iflags(2)**.)

In addition, the errors documented in **stat(2)** can also occur.

ERANGE

The size of *name* or *value* exceeds a filesystem-specific limit.

STANDARDS

Linux.

HISTORY

Linux 2.4, glibc 2.3.

SEE ALSO

getfattr(1), **setfattr(1)**, **getxattr(2)**, **listxattr(2)**, **open(2)**, **removexattr(2)**, **stat(2)**, **symlink(7)**, **xattr(7)**